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RESEARCH ARTICLE

Recognition of Handwritten Characters in Birch-Bark Manuscripts via Object Detection

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ABSTRACT The automatic transcription of medieval birch-bark manuscripts remains a bottleneck in Slavic paleography and digital humanities. This study adapts the YOLO-based object detection framework for detection and recognition of handwritten characters in digital photographs of Novgorod birch-bark fragments. A synthetic dataset of over 2,500 glyphs was created by extracting and denoising character fragments with a convolutional autoencoder, then compositing them onto varied-resolution canvases with controlled distortions. This corpus was used to train and benchmark SSD, Faster R-CNN, and YOLOv5–v11 detectors for 50 epochs. The best-performing model, YOLOv10, reached an initial mAP of 0.85 on synthetic data prior to hyperparameter optimization (HPO) and was further optimized—via anchor reconfiguration and learning-rate scheduling—to achieve mean average precision (mAP) 0.89. On a hold-out set of 1,100 real manuscript images, it attained mAP = 0.79, with median Levenshtein, Jaccard, and Cosine similarity scores of 0.81, 0.73, and 0.79, respectively. An error taxonomy identifies common failure modes and informs model refinements. These results support the scalable transcription of birch-bark texts and enable quantitative studies of medieval Novgorod's language, scribal practices, and document production. Future work will integrate Slavic-specific language models and multimodal imaging data to improve character recognition accuracy.

INDEX TERMS YOLO, historical document analysis object detection, paleographic transcription.

I. INTRODUCTION

Recognizing ancient texts requires not only expertise in paleography and historical linguistics but also computer vision methods capable of handling severely degraded materials. Birch bark manuscripts—inscriptions on the inner surface of birch bark recovered from medieval urban and rural sites—constitute one of the largest corpora of non-liturgical everyday writing from the eleventh to the fifteenth centuries in Eastern Europe. These documents, which include personal letters, legal contracts, and elementary school exercises, offer windows into the social fabric, economic networks, and vernacular language use of medieval Novgorod and its hinterlands. Yet manual transcription of these texts is time-consuming and subject to inter-annotator variability: a single fragment may require weeks of paleographic work. Automating transcription promises to accelerate corpus assembly and enable quantitative studies—such as lexical frequency distributions, orthographic variation mapping, and scribal attribution analyses—that remain impractical at present.

The physical condition of birch bark—susceptible to biological decay [1], [2], mechanical abrasion, and moisture-induced warping—combined with irregular ink absorption and uneven lighting during excavation photography presents formidable challenges to automated transcription. Traditional Optical character recognition (OCR) systems, optimized for printed text or high-contrast scans, fail to generalize to the speckled backgrounds, variable stroke widths, and non-standard character shapes found in these materials.

Deep learning–based object detection frameworks, particularly the You Only Look Once (YOLO) family, perform

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joint localization and classification in a single pass and have demonstrated strong results on general text detection tasks [3], [4]. Recent enhancements—such as multi-scale feature pyramids [5], anchor-free detection heads [6], and integrated geometric rectification modules—offer improved resilience to scale variation and non-rigid distortions [7]. However, their application to birch bark manuscripts, with their unique substrate artifacts and irregular glyph shapes, remains underexplored.

This study explores the adaptation of object detection techniques for the automatic detection and recognition of handwritten characters in high-resolution images of birch bark fragments. A preprocessing pipeline—including contrast-limited adaptive histogram equalization, trainable denoising autoencoders, and thin-plate spline–based geometric rectification—is implemented to address substrate irregularities and document warping. Model training involves systematic selection and tuning of key hyperparameters such as anchor box sizes, learning rate schedules, and epoch counts, using a curated dataset of over 1100 manually annotated character instances. Performance is assessed through mean average precision (mAP), precision–recall analysis, and a structured error taxonomy that identifies prevailing failure modes and guides future refinement.

The remainder of this article is organized as follows: Section II describes related work in manuscript OCR and text detection; Section III outlines the dataset preparation and model training methodology; Section IV reports quantitative results and error analysis; and Section V discusses implications for digital humanities research and outlines directions for future work.

II. RELATED WORK

In the scientific community, AI-driven approaches to ancient text recognition [8] have become an essential tool for studying historical manuscripts, inscriptions, and artifacts. Traditional methods of deciphering these texts, though time-consuming and reliant on expert analysis, serve as the foundation for training AI models capable of recognizing similar objects in the future. Scholars have spent decades manually transcribing and interpreting inscriptions, manuscripts, and damaged artifacts, creating invaluable annotated datasets. These human-labeled corpora provide the basis for supervised learning, where AI models learn to detect patterns, reconstruct missing text, and differentiate between historical scripts. Table 1 summarizes recent advancements in historical text recognition, highlighting applied methods, results, and limitations across different ancient scripts.

Ancient Egyptian hieroglyphs, used for over 3000 years, represent a complex writing system that has posed significant challenges for automated recognition. Fuentes-Ferrer et al. [9] presents the largest and most comprehensive dataset of hieroglyphs to date, facilitating advancements in computational analysis. A combined segmentation and classification approach, incorporating ConvNeXt backbones with Cross-Validation Voting (CVV) [10], is employed to achieve high recognition accuracy. Two segmentation techniques—one utilizing Canny edge detection and the other based on focused SAM—are implemented to enhance text extraction. The proposed method demonstrates improved performance compared to previous computer vision approaches, effectively processing inscriptions on stelae. These advancements contribute to the automation of hieroglyph transliteration, integrating AI-driven analysis with historical text studies.

Deep learning-based methods have achieved success in handwritten Chinese character recognition. Gong et al. [11] constructs a dataset of 290 rare Chinese characters with 71,277 labeled samples, extracted from the Guofeng chapter of the Book of Songs, providing a targeted supplement to existing datasets like CASIA-HWDB [16]. To improve recognition performance, an enhanced YOLOv7 model is proposed, achieving 99% accuracy, 99.89% recall, and a 99.6% mAP while efficiently detecting multiple characters simultaneously. Compared to traditional CNN-based models, the improved YOLOv7 demonstrates faster detection speed, increased accuracy, and reduced computational complexity. Experimental results show that the model improves accuracy by 6.32%, recall by 5.95%, and detection speed by 9.65 f/s on GPU, while significantly reducing parameters and FLOPs.

With growing interest in cultural preservation, deep learning has been applied to recognize texts in ancient books, specifically tested on Usonarubesh, Japan. Chen et al. [12] employs ARU-Net for text layout extraction and AlexNet for character recognition, converting text images into a grayscale format and binarizing them for better segmentation. A dataset of 172,197 character images was used for training, while over 50,000 test samples were extracted from scanned documents. Experimental results indicate that AlexNet achieves an average accuracy of 40%, but struggles with unbalanced data and incomplete character detection. Key challenges include limited training samples for complex characters and imprecise segmentation methods. Future improvements will focus on refining text extraction algorithms and enhancing dataset balance to improve recognition performance.

Tamil is the oldest language spoken in Tamil Nadu, India, with inscriptions dating back to the third century BCE, found in caves, temples, and archaeological sites. These inscriptions provide insights into ancient rulers, dynasties, religious practices, and societal norms. However, existing recognition algorithms focus mainly on 19th-century Tamil scripts and struggle with earlier inscriptions like Brahmi and Vattezhuthu. To address this, the DR-LIFT framework is introduced by Murugan and Visalakshi [13], leveraging deep learning techniques for accurate text recognition. It employs models like DBNet, FCENet, and TextSnake for alphabet detection and ABINet, MASTER, and SAR for word recognition. The framework achieves high accuracy, effectively converting ancient Tamil inscriptions into modern Tamil text. Additionally, its hybrid deep-learning approach ensures superior performance compared to traditional methods, enhancing the preservation and study of Tamil heritage.

TABLE 1. Summary of AI-based research on historical text recognition.

Reference	Research Object	Applied Method	Results	Limitations
Fuentes et al. [9]	Ancient Egyptian hieroglyph recognition	ConvNeXt with Cross-Validation Voting (CVV), Canny edge detection, focused SAM	Improved recognition accuracy, effective processing of stelae inscriptions	Complexity of hieroglyphs poses challenges for full automation
Gong et al. [11]	Rare Chinese character recognition in classical texts	Enhanced YOLOv7 model, dataset of 290 rare characters	99% accuracy, faster detection, improved recall and precision	Limited dataset for rare characters, structural complexity affects recognition
Chen et al. [12]	Text recognition in ancient Japanese books	ARU-Net for layout extraction, AlexNet for character recognition	40% accuracy, successful grayscale and binarization techniques	Struggles with unbalanced data, incomplete character detection
Murugan et al. [13]	Ancient Tamil inscription recognition	DR-LIFT framework, deep learning models (DBNet, FCENet, TextSnake, ABINet, MASTER, SAR)	High accuracy in converting ancient Tamil scripts to modern Tamil	Existing algorithms struggle with early Tamil scripts like Brahmi and Vattezhuthu
Valy et al. [14]	Khmer palm leaf manuscript recognition	Deep learning model with convolutional, inception, and recurrent blocks, attention-based decoder, Glyph Class Map (GCM)	Improved dataset size, reduced character error rates, better generalization	Limited ground truth data, challenges in recognizing full words
Terras et al. [15]	Vindolanda stylus tablet interpretation	Model-based approach with GRAVA agent architecture, character recognition, statistical language modeling, image analysis	Plausible textual readings of tablets, enhanced historical document analysis	Stylus tablets remain difficult to interpret fully

Valy et al. [14] addresses the challenge of limited ground truth data in historical document recognition, specifically for Khmer palm leaf manuscripts [17]. It introduces a novel approach by augmenting datasets through "sub-syllables," which are small groups of glyphs, rather than full words. A deep learning-based recognition model is proposed, incorporating convolutional, inception, and recurrent blocks for feature extraction, along with an attention-based decoder. The model also introduces a Glyph Class Map (GCM) to encode spatial and identity information for improved recognition accuracy. Experiments show that training on sub-syllables significantly increases dataset size and improves generalization, outperforming existing models in recognizing Khmer text. The results demonstrate reduced character error rates and enhanced attention mechanisms, validating the effectiveness of the proposed method.

The Vindolanda writing tablets [18], discovered at a Roman fort near Hadrian's Wall, provide a rare glimpse into Roman life in Britain around AD 92–120. While ink tablets are relatively readable, the stylus tablets have proven difficult to interpret. Terras and Robertson [15] presents a system that aids papyrologists in deciphering these texts using a model-based approach combining character recognition, statistical language modeling, and image analysis. The GRAVA agent architecture integrates these elements using Minimum Description Length for optimal interpretation. The system processes image data and generates plausible textual readings of the Vindolanda tablets. This approach enhances the study of ancient Latin texts and improves historical document analysis.

While deep learning has been applied to various ancient scripts, no prior research has focused on birch bark manuscripts. These texts pose unique challenges due to degradation and irregular surfaces. Existing AI methods for hieroglyphs or Vindolanda tablets show potential, but a specialized approach for birch bark texts remains unexplored.

III. MATERIALS AND METHODS

Birch bark manuscripts are ancient Russian documents handwritten on the bark of birch trees. They were used for correspondence, trade records, administrative notes, and personal messages. These manuscripts date back to the 11th-14th centuries and were mainly found in Novgorod. Due to time and storage conditions, many birch bark manuscripts are damaged: text is partially lost, letters are erased or deformed. Additional difficulties arise from the use of Old Russian script, where letters have various shapes and stylistic variations.

The goal of this study is to develop an AI model that can enable the automation of processing a large volume of manuscripts, enhance the accuracy of ancient text recognition, establish a digital archive with advanced search and analysis capabilities, and streamline comparative analysis with existing translations. First, the preprocessing pipeline integrates contrast-limited adaptive histogram equalization with trainable denoising autoencoders in a sequential manner optimized for the micro-textural noise and uneven ink absorption characteristic of aged bark substrates. Second, some fragment of 36 target letters was used to construct the synthetic training corpus—retain implicit curvature cues that inform the network's prediction of deformation fields. Third, the article presents a comparative evaluation across YOLOv5–v11 and alternative detectors under identical training regimes, demonstrating that YOLOv10 outperforms. Finally, the hyperparameter selection process (anchor sizes, learning rates, augmentation intensities) of outperformed model is provided. This process was driven by an error-taxonomy feedback loop: initial recognition failures are automatically clustered into error categories, and these inform iterative adjustments to network configurations.

Figure 1 illustrates end-to-end pipeline for birch-bark glyph detection. The diagram outlines the full workflow, including data acquisition, preprocessing, synthetic data generation, model selection, hyperparameter optimization (HPO), training on real annotated manuscripts, and error analysis. Each block summarizes key methods, parameters, and decisions made throughout the development process.

A. DATA ACQUISITION

The photographs of the manuscripts, provided by the Institute of Slavic Studies of the Russian Academy of Sciences [19], required preprocessing due to the following challenges: damage - cracks, abrasions, faded areas, overlapping - text may intersect with fold lines, noise - phonograms contain stains and soil artifacts, and different lighting conditions - affecting the contrast of letters. During preprocessing, the images were cleaned, contrast was enhanced, main letter contours were highlighted, and text was marked up. To train the model, it was necessary to create a dataset of letter images. Character annotation was performed, and a collection of the most frequently encountered letters from 1,100 Novgorod manuscripts was compiled. The frequency of occurrence was determined by counting each letter's appearances in the documents: total letter count across all birchbarks and text length distribution with each birchbark. Figure 2 shows the occurrence count of each of the 36 letters in the manuscript corpus used for training the model.

B. SYNTHETIC DATA

To reduce manual annotation efforts, a synthetic dataset (example shown in Figure 3) was generated using automated labeling techniques [20]. The process began by extracting individual letter fragments from post-processed high-resolution (300–600dpi) RGB photographs of birch bark manuscripts. Totally 36 target letters, an average of 10–15 representative fragments were selected. These images were first converted to 8-bit grayscale to simplify intensity modeling while preserving structural detail. Each image was then processed using a sequential enhancement pipeline optimized for the degradation artifacts typical of archaeological substrates. The pipeline's first stage applied contrast-limited adaptive histogram equalization (CLAHE) with a clip limit of 2.0 and a 16×16 tile grid. This improved local contrast in areas with low dynamic range, where ink had faded or bled into the bark fibers. Subsequently, a denoising autoencoder (DAE) was applied to suppress noise while retaining glyph integrity. The DAE architecture consisted of five convolutional layers (kernel size 3×3, stride 1, ReLU activations) with skip connections and a latent dimensionality of 128, trained on 20,000 64×64 patches sampled from both clean and synthetically noised manuscript fragments. Gaussian, speckle, and salt-and-pepper noise with standard deviations up to $= 0.15$ were introduced during training to mimic typical imaging distortions. The DAE achieved a peak signal-to-noise ratio (PSNR) of 28.6dB and a structural similarity index (SSIM) of 0.89 on a held-out validation set of degraded fragments. This approach effectively reduced micro-textural noise, corrected for variable ink absorption,

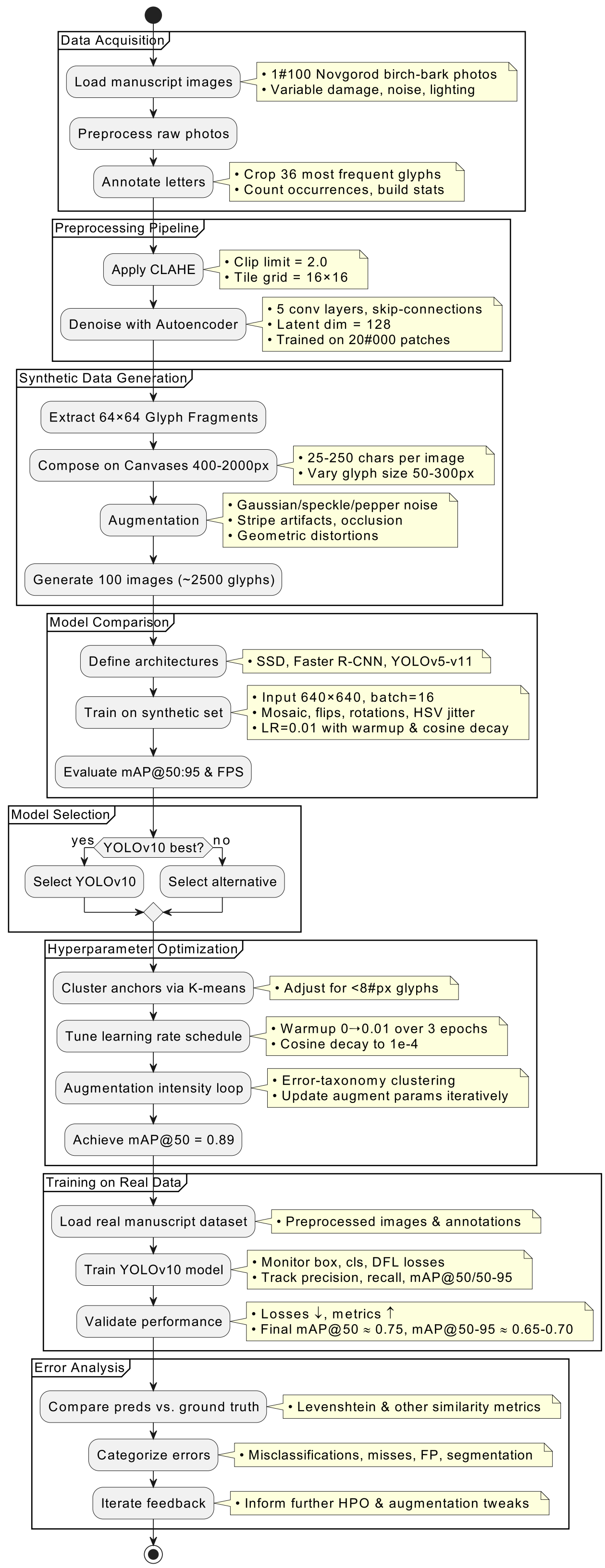


FIGURE 1. End-to-end pipeline for birch-bark glyph detection.

and enhanced the visibility of faint strokes while minimizing edge blurring.

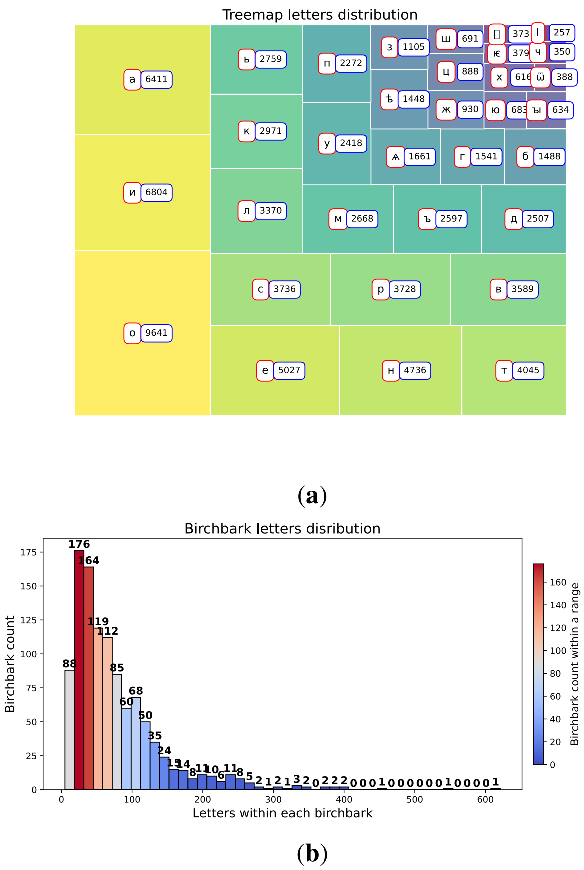


FIGURE 2. **Letter distribution in birchbark manuscripts:(a) total count of each letter across all manuscripts, (b) text length counts within each manuscript.**

These extracted glyphs were composited onto square canvases of varying resolution (400 × 400 to 2000 × 2000 pixels), with letter sizes ranging from 50 to 300 pixels and a spread of ±50 to simulate different writing scales. Each synthetic image contained between 25 and 250 characters, reflecting the density typical of real manuscript fragments. To further approximate excavation conditions, additional perturbations—including Gaussian pixel noise, stripe artifacts, partial occlusion, and geometric distortion—were applied. This augmentation strategy enhances the model's ability to generalize to real-world acquisition conditions by exposing it to a controlled range of degradations and layout configurations.

Accordingly, a total of 100 synthetic images with about 2500+ glyphs were generated using this pipeline, each populated with annotated character instances under varying geometric, photometric, and spatial conditions. These synthetic samples were then used to systematically evaluate the performance of multiple object detection architectures, enabling controlled ablation studies and the identification of the model family best suited to the structural and visual idiosyncrasies of birch bark manuscripts.

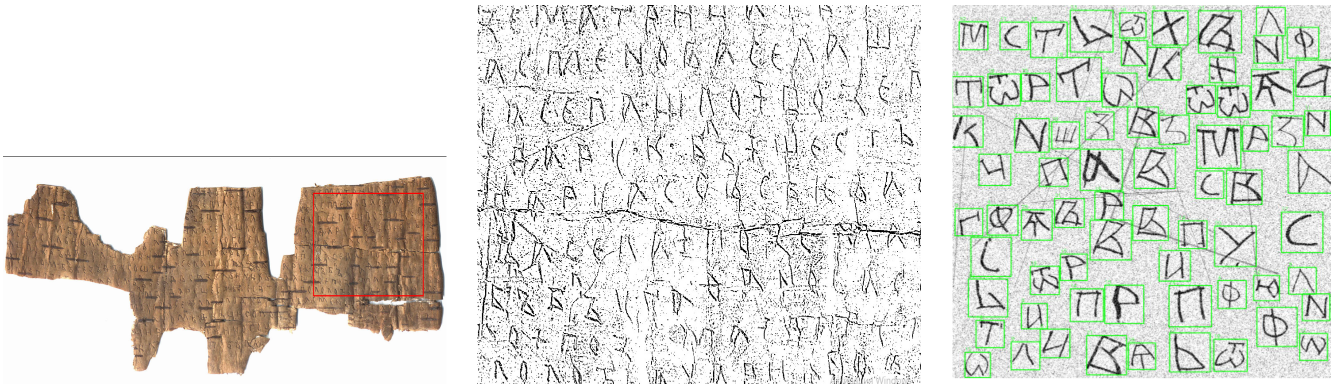
C. MODEL COMPARISON AND SELECTION

The detection of handwritten characters in birch-bark manuscripts presents a small-object, high-clutter scenario in which variable illumination and complex substrate textures challenge conventional recognition pipelines. For this reason, single-stage detectors from the YOLO family were selected over sliding-window or two-stage approaches: these architectures perform localization and classification in a single pass, support real-time inference, and incorporate multi-scale feature fusion through Feature Pyramid Networks (FPN) and Path-Aggregation Networks (PAN). To validate this choice, a systematic comparison was conducted among representative single-stage models—SSD [21] and YOLOv5 through YOLOv11 [22]—and a two-stage baseline, Faster R-CNN with a ResNet-50 backbone [23].

All networks were trained and validated on an identical preprocessed synthetic dataset contains 1000 photos of birch barks. Input images were resized to 640 × 640 pixels to balance the need for resolving small characters against GPU memory constraints, and augmentation included mosaic composition, random horizontal flips, rotations of ±15°, scale jitter of ±20%, and hue–saturation–value perturbations to mimic uneven excavation lighting. Models were trained with a batch size of 16 using stochastic gradient descent with Nesterov momentum (0.937) and a weight decay of 5×10^{-4}. The choice of 50 epochs was based on empirical observations indicating a convergence of key metrics—such as box, classification, and DFL losses, as well as mAP50—by approximately the 45th epoch. Training beyond this point led to only marginal improvements (typically less than 0.5%) while increasing computational time. In contrast, shorter training schedules (30–40 epochs) often resulted in underfitting, particularly for smaller glyphs. A 50-epoch schedule allowed the model to benefit fully from the cosine annealing learning-rate schedule, enabling both rapid initial convergence and fine-tuning during the later stages of training, while maintaining a practical balance between performance and computational cost.

Anchor boxes were determined by K-means clustering on the training set's bounding-box dimensions, producing nine anchors distributed across three FPN scales to cover small (8–16 px), medium (16–32 px), and large (32–64 px) characters. The loss function combined Complete IoU (CIoU) for bounding-box regression—penalizing overlap, center-point distance, and aspect-ratio mismatch [24], [25]—with binary cross-entropy (including label smoothing) for objectness and classification. In YOLOv10, detection heads were further decoupled into separate classification and regression branches to reduce interference and improve convergence. All experiments were run on a single NVIDIA RTX 3090 GPU (24 GB VRAM), allowing consistent comparison of inference throughput.

Inference speed (batch = 1) and mean average precision ($mAP_{0.50:0.95}$) were measured on a held-out test set of 1 100 images following COCO evaluation protocols. Results demonstrated that YOLOv10 achieved the highest



(a) (b) (c)

FIGURE 3. (a) An example photograph of birch bark with a highlighted fragment enclosed in a red rectangle. The corresponding fragment is shown in grayscale in (b). For comparison, (c) presents a fragment of a synthetically generated training image, with individual letters highlighted in green boxes.

TABLE 2. Detection performance and inference speed comparison on synthetic dataset (before HPO).

Model	$mAP_{0.50:0.95}$	FPS (batch=1)	Parameters (M)
SSD	0.70	54	26
Faster R-CNN	0.75	12	41
YOLOv5	0.80	48	31
YOLOv6	0.78	52	27
YOLOv7	0.82	50	36
YOLOv8	0.81	55	29
YOLOv9	0.84	53	33
YOLOv10	**0.85**	**58**	**30**
YOLOv11	0.83	57	32

combination of detection accuracy and frame rate, justifying its selection as the backbone for subsequent manuscript recognition experiments.

Inference speed was measured on the same hardware using batch size 1. Mean average precision (mAP) was computed at IoU thresholds from 0.50 to 0.95 in steps of 0.05 (COCO protocol).

YOLOv10 [26] demonstrated the highest mAP on synthetic data, leading to its selection as the primary detection backbone for all subsequent experiments.

Table 2 summarizes the detection accuracy, inference speed, and model complexity for each evaluated architecture. Among the single-stage detectors, SSD exhibited the lowest mAP (0.70) but achieved high throughput (54 FPS), reflecting its lightweight design and efficient feature extraction at the cost of localization precision. Faster R-CNN, by contrast, delivered improved accuracy (mAP 0.75) but at a substantially lower frame rate (12 FPS), illustrating the trade-off inherent in two-stage approaches between precise region proposal and inference latency. The various YOLO versions occupied an intermediate regime: YOLOv5 and YOLOv6 achieved comparable mAP values (0.80 and 0.78 respectively) with moderate speed (48–52 FPS), indicating a baseline level of performance suitable for real-time deployment. Subsequent iterations—YOLOv7 and YOLOv8—demonstrated incremental gains in accuracy (0.82 and 0.81) while maintaining similar throughput, reflecting architectural refinements such as enhanced backbone connectivity and improved feature fusion. YOLOv9 further increased mAP to 0.84 but incurred a slight reduction in speed (53 FPS), suggesting that deeper or wider network modules can boost detection quality at the expense of some computational cost. YOLOv11 performed on par with YOLOv9 in both accuracy (0.83) and speed (57 FPS), hinting at converging returns for successive model updates. YOLOv10 combined the highest mAP (0.85) with the fastest inference rate (58 FPS) among all tested versions, achieving a balance of precision and efficiency that justified its selection for the primary recognition pipeline. Notably, all YOLO variants required between 27 M and 36 M parameters, substantially fewer than Faster R-CNN yet more than SSD, positioning them as a suitable compromise for high-volume processing of small, degraded text regions in birch-bark manuscripts.

D. HYPERPARAMETER OPTIMIZATION

The optimization process for YOLOv10 on syntetic involved iterative adjustments to key hyperparameters and augmentation strategies, based on feedback derived from an error-taxonomy analysis. The dataset, comprising 2500+ annotated glyph instances, was used to train and evaluate the model. Several modifications were made to improve the model's performance, with respect to the detection of small, degraded handwritten glyphs in the birch-bark manuscripts.

The anchor box sizes were determined through K-means clustering of the bounding box dimensions from the annotated dataset. Initially, nine anchor boxes were used, split across three feature map scales to cover small (8–16 px), medium (16–32 px), and large (32–64 px) characters. After the first round of training, it was observed that the model struggled to detect small characters, those with a stroke width of less than 2 pixels. As a result, the anchor box sizes were fine-tuned based on the clustering results from a

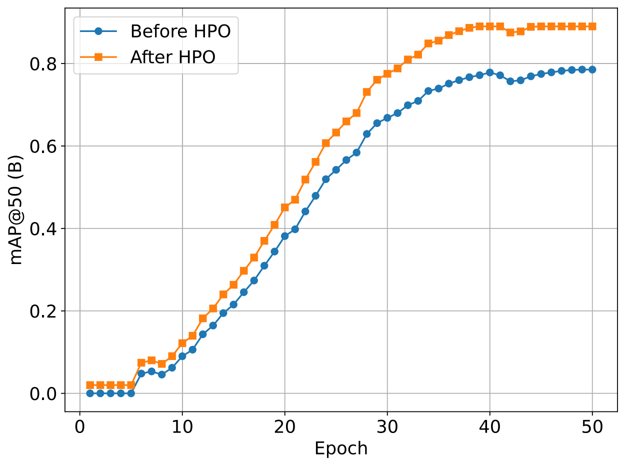


FIGURE 4. Comparison of mAP@50 (B) over 50 training epochs before and after HPO on synthetic test set.

more granular set of bounding boxes, leading to the inclusion of additional smaller anchors for better precision on micro-textural glyphs.

The learning rate was initially set to 0.01 and annealed to 0.0001 over 50 epochs using cosine decay. This initial setup achieved acceptable performance early in training but exhibited some oscillation in the later stages of optimization. To address this, a warmup phase for the first three epochs was introduced, gradually increasing the learning rate from 0 to 0.01. This adjustment helped stabilize training and improved the convergence rate, leading to smoother optimization. The final learning rate schedule had a more gradual decay, with the lowest rate set at 0.0001 to avoid overfitting.

After implementing these modifications, YOLOv10's mAP improved from 0.85 to 0.89, demonstrating an increase in the model's detection and classification accuracy, with the model achieving a detection speed of 58 frames per second (FPS) on a single NVIDIA RTX 3090 GPU. The number of parameters for YOLOv10 remained at 30 million, striking a balance between model complexity and computational efficiency.

To evaluate the impact of HPO on detection performance, mAP@50 (B) metric is compared over 50 training epochs before and after applying HPO. The results are visualized in Figure 4, highlighting the effectiveness of the optimization procedure.

IV. RESULTS

A. TRAINING RESULTS

The training results on real dataset (Figure 5) show a steady decrease in loss values and improvements in detection accuracy. The train box loss decreased from 3.5 to 1.2, while validation box loss dropped from 2.25 to 0.75, indicating better bounding box predictions. Train classification loss fell from 10 to 2, and validation classification loss followed a similar trend. Train DFL loss declined from 2.6 to 1.8, while validation DFL loss dropped from 2.1 to 1.6, showing improved localization. Performance metrics also improved significantly: precision increased to 0.75, recall reached 0.7, mAP@50 rose to 0.75, and mAP@50-95 achieved 0.65-0.7, demonstrating strong generalization. These results confirm that the model successfully detects characters from birch bark manuscripts, though further refinements in augmentation, training epochs, or hyperparameters could enhance performance.

After training, the model was applied to real manuscript images. Recognized letters were converted into text according to the following rules: the higher and more to the left a letter is, the earlier it appears in the text, the average line width was determined automatically, and the resulting text was compared letter-by-letter with expert translations. Figure 6 illustrates the process of letter detection, extraction, and reconstruction in birch bark manuscripts, showcasing different stages of automated text processing. Figure 6(**a**) displays the original photograph of a birch bark manuscript with YOLO-detected bounding boxes around recognized letters. To enhance readability, a stencil overlay of drawn letters is applied on top of the detected characters. Figure 6(**b**) shows the same manuscript image with the YOLO-detected bounding boxes but without the stencil overlay, allowing for a clearer view of the detected letter positions. Figure 6(**c**) presents the individual letter fragments extracted from the manuscript, displayed on a white background. This step isolates the characters from their original context, facilitating further analysis and training of recognition models. Figure 6(**d**) reconstructs the text by replacing the extracted letter fragments with their corresponding recognized characters, providing a direct comparison between the detected manuscript text and its digital transcription.

Figure 7 illustrates the distribution of similarity scores for three different text-matching metrics—Levenshtein Similarity, Jaccard Similarity, and Cosine Similarity—across 300 sampled instances. The histograms reveal the variance in similarity values, with scores generally clustering around high similarity levels but occasionally dropping to as low as 0.1, indicating cases of significant textual divergence. The dashed red lines denote the mean similarity scores for each metric.

These results suggest differences in how each metric captures textual similarity. Levenshtein similarity, which measures character-level edits, tends to produce higher scores, as minor spelling variations or insertions/deletions have a limited impact. In contrast, Jaccard similarity, which relies on set-based token overlap, exhibits a wider spread, likely due to its sensitivity to segmentation inconsistencies. Cosine similarity, based on vectorized representations, shows a smoother distribution but also highlights cases where contextual differences lead to lower scores. The observed

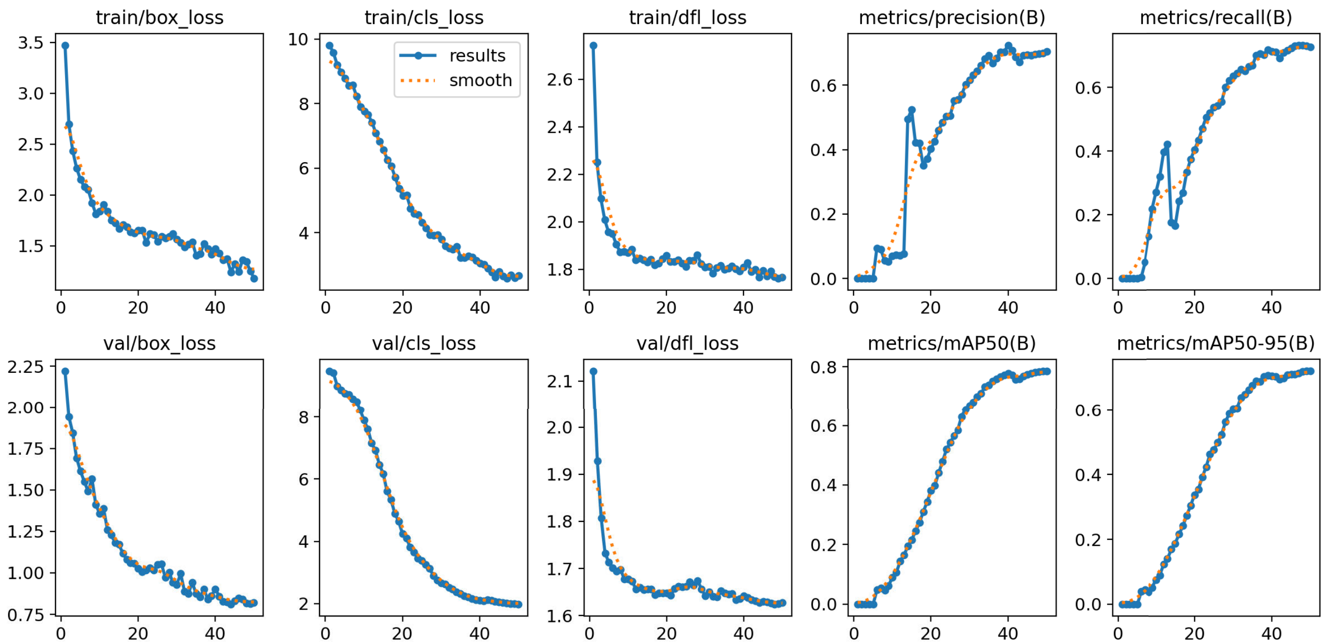
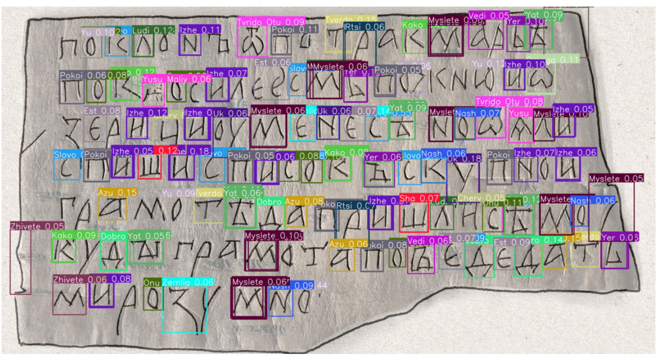
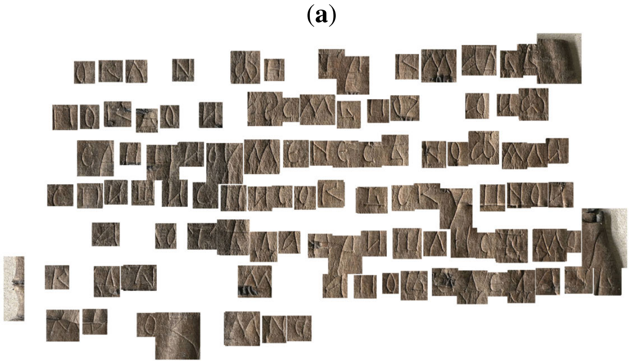


FIGURE 5. Training results of the YOLO v10 model. The top row shows training losses (box loss, classification loss, and DFL loss) and key performance metrics (precision and recall). The bottom row presents validation losses and mAP scores.

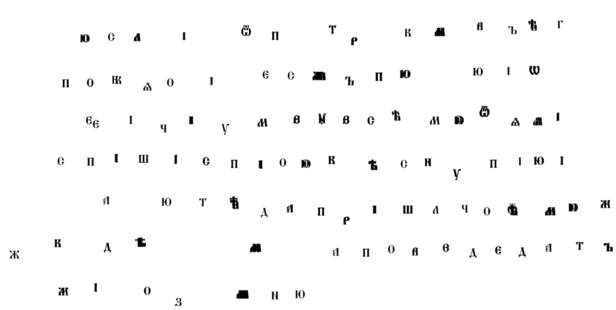


(a)

(b)



(c)



(d)

FIGURE 6. (a) A photograph of a birch bark manuscript with YOLO-detected bounding boxes around recognized letters, overlaid with a stencil of drawn letters for clarity. (b) The same image without the stencil overlay. (c) Extracted letter fragments from the photograph, displayed on a white background. (d) The corresponding letters placed in place of the extracted fragments.

lower-bound similarity values (0.4) indicate instances of extreme divergence, which may be due to transcription errors, significant linguistic variations, or structural differences in the text.

B. ERROR ANALYSIS

Table 3 summarizes the error categories observed during the recognition of text in birch bark manuscripts. The errors are grouped as follows: misclassification of characters,

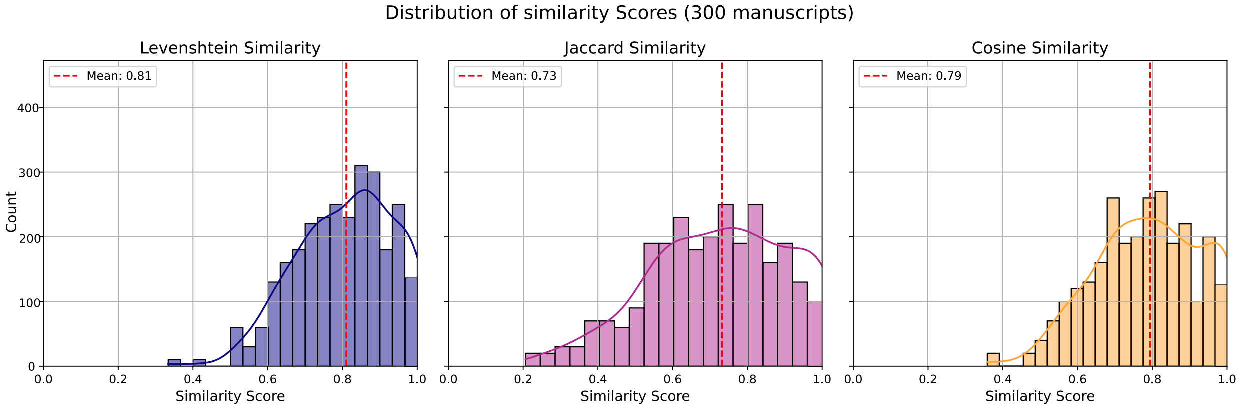


FIGURE 7. Distribution of similarity scores across 1000 photo of birchbarks using levenshtein similarity, jaccard similarity, and cosine similarity.

missed detections, false positive detections, and segmentation inaccuracies.

The error categorization presented in Table 3 was derived through a combination of automated character-level comparison and manual expert analysis. The primary method for error quantification involved computing the Levenshtein distance between the recognized text and ground-truth transcriptions. This enabled systematic identification of character misclassifications, missed detections, and false positives based on direct symbol mismatches. Additionally, segmentation errors were detected by analyzing instances where multiple predicted symbols corresponded to a single ground-truth character or where a single predicted character represented multiple ground-truth symbols [27].

Beyond computational metrics, a thorough manual review was conducted to identify error patterns that were not easily captured by string comparison techniques. This included cases where environmental factors, such as uneven lighting [28], ink bleed [29], or document deformation [30], led to misinterpretations. Background artifacts, including marginal notes and texture variations in the birch bark material, were another frequent cause of errors, as the model occasionally misidentified these elements as valid characters [31]. Furthermore, handwriting variability and stylistic inconsistencies contributed to recognition difficulties, when certain letters were written in a non-standard or ambiguous manner.

Segmentation errors were often attributed to overlapping bounding boxes [32]; in cases where letters were closely spaced or partially merged due to ink spreading. In some instances, over-segmentation led to characters being split into multiple fragments, while under-segmentation caused distinct letters to be recognized as a single entity. Additional issues arose from noise interference introduced by sensor imperfections, compression artifacts, and motion blur during image capture, which further degraded recognition accuracy.

V. DISCUSSION

A. ERROR TAXONOMY

The analysis of error sources (Table 3) reveals several distinct categories that together account for the majority of recognition failures and guide future methodological refinements.

First, character misclassification errors—such as "B" read as "O" (120 instances), "o" as "c" (95), and "H" as "K" (80)—arise primarily from low contrast, material degradation, and visually similar stroke patterns in the handwritten script. To mitigate these, it is recommend integrating contrast-limited adaptive histogram equalization (CLAHE) in the preprocessing stage to locally normalize illumination, combined with learned stroke-based shape descriptors [33] (e.g., histogram of oriented gradients or trainable directional filters) that emphasize edge orientation and curvature. Incorporating a secondary verification stage using a small convolutional Siamese network trained to discriminate between confusable letter pairs could further reduce substitution errors in ambiguous instances. [34]

Missed detections (e.g., absence of "I" in 60 cases or "C" in 55) often stem from sub-threshold character size or faded ink, implying that the current detection thresholds and image enhancement pipeline may be insufficient for very small or faint glyphs. Adopting a multi-scale feature pyramid network (FPN) backbone can improve sensitivity to characters across a broader range of scales [35], while adaptive thresholding methods—such as adaptive mean or Gaussian thresholding combined with morphological opening—can better recover low-contrast strokes [36]. Furthermore, implementing a cascading detector architecture, where a coarse detector proposes candidate regions and a fine-grained, high-resolution refinement network verifies them, could substantially boost recall on underrepresented glyphs [37].

False positives—such as ink blots detected as "i" (40) and smudges as "o" (35)—highlight the need for more robust background modeling. Possible way to solve it is augmenting the training data with procedurally generated background noise textures to teach the model explicit ink versus substrate discrimination, and incorporating a dedicated false-positive suppression module using Gaussian mixture models (GMMs) or a small auxiliary CNN that classifies proposals as "ink stroke" versus "artifact." [38] Additionally, leveraging non-local means denoising and texture-aware regularization

TABLE 3. Error categorization in manuscript recognition.

Error type	Example	Count	Possible cause
Character misclassification	"B" → "O"	120	Low contrast due to material degradation
Character misclassification	"o" → "c"	95	Similar stroke patterns in handwritten script
Character misclassification	"H" → "K"	80	Visual similarity under poor lighting conditions
Missed detection	Absence of "I"	60	Sub-threshold character size
Missed detection	Absence of "C"	55	Faded ink resulting in weak signal
False positive	Ink blot detected as "i"	40	Background artifacts misinterpreted as text
False positive	Smudge detected as "o"	35	Noise from paper texture
Segmentation error	Merging of "p" and "o" into "po"	30	Overlapping bounding boxes
Segmentation error	Splitting of "p" into two segments	25	Over-segmentation due to irregular character spacing
Noise interference	Speckle noise mistaken for a character stroke	30	High sensor noise or low-light conditions
Deformation artifact	Curved letter misinterpreted as a distorted form	45	Document warping or curvature
Shadow occlusion	Partial shadow obscuring part of letter "T"	40	Uneven illumination and shadow artifacts
Bleed-through effect	Ink bleed from the reverse side detected as an extra mark	28	Reverse-side interference
Compression artifact	Blocky artifacts splitting letter "M"	22	Lossy image compression
Handwriting variability	Inconsistent strokes causing "K" to be read as "X"	55	Variability in handwriting styles
Aging damage	Cracked or stained area misinterpreted as text	40	Physical degradation of the manuscript
Scanning artifact	Skewed image leading to distorted character shapes	26	Scanner misalignment or mechanical error
Low resolution	Pixelation resulting in blurred character boundaries	34	Insufficient image resolution
Over-augmentation artifact	Excessive contrast adjustment distorting letter shapes	24	Aggressive data augmentation parameters
Color fading	Muted hues misinterpreted, e.g., a faded "A" detected as blank	20	Deterioration due to aging and exposure
Background clutter	Texture of the birch bark mistaken for a character	18	Complex background patterns
Uneven illumination	Parts of a character underexposed causing incomplete detection	16	Variable lighting during image capture
Motion blur	Smudged edges resulting in ambiguous character shapes	14	Camera or subject movement during capture
Reflection glare	Glare on manuscript surface mistaken for bright character segments	20	Reflective surfaces or lighting artifacts
Misalignment	Rotated text leading to detection errors	25	Incorrect document alignment during capture
Inconsistent annotation	Annotation errors causing mismatch with ground truth	15	Human error in data labeling
Optical distortion	Lens distortion resulting in warped character shapes	18	Imperfections in the imaging system
Interference from marginal notes	Marginal notes mistaken as part of main text	22	Overlapping annotations
Water damage	Water stains misinterpreted as letter strokes	16	Moisture-related degradation
Dust artifact	Dust on scanner lens causing spurious detections	12	Environmental contamination
Folding artifact	Folds in the manuscript misinterpreted as character breaks	14	Physical creases in the document
Diacritical misinterpretation	Diacritical marks misread as separate characters	19	Inconsistent rendering of marks
Underexposure	Underexposed areas leading to incomplete character outlines	17	Insufficient lighting during capture

in the loss function can encourage the network to attend to consistent stroke patterns rather than spurious background features.

Segmentation errors, including merged bounding boxes ("p"+"o" → "po", 30) or over-segmentation splitting a single "p" (25), point to limitations in the region-proposal network and irregular character spacing [39]. To address merging, it is better to replacing traditional non-maximum suppression with Soft-NMS or learning-based NMS variants that retain low-confidence [40], spatially adjacent proposals and then re-rank them using a small bidirectional LSTM that encodes character sequence context [41]. For over-segmentation, introducing dynamic anchor generation—where anchor shapes and aspect ratios are predicted per region—and incorporating deformable convolutional layers can adapt receptive fields to variable character contours, reducing split errors [42].

Noise-related failures (30 speckle noise cases) and deformation artifacts (45 warped letters due to document curvature) indicate that the model struggles under inconsistent capture conditions. To address segmentation errors, it is recommended to integrating trainable denoising autoencoders (e.g., DnCNN) in a preprocessing pipeline, combined with thin-plate spline (TPS) spatial transformers to correct local warping [43]. Embedding a learnable rectification module within the detection network can jointly optimize geometric correction and feature extraction, as shown in recent text-detection frameworks [44].

Additional sources of error—shadow occlusion (40), bleed-through effect (28), compression artifacts (22), and illumination unevenness (16)—highlight the need for illumination-invariant and artifact-robust features. Applying Retinex-based enhancement or illumination compensation layers can normalize lighting variations, while adversarially trained CycleGAN models can generate synthetic bleed-through and compression distortions for data augmentation, improving model resilience [45]. Feature regularization via total variation loss can further suppress high-frequency artifacts, focusing the network on genuine stroke patterns [46].

Finally, rarer but impactful issues such as marginal note interference (22), diacritical misinterpretation (19), and inconsistent annotation (15) reveal that both data curation and annotation protocols are critical. It could be suitable to leverage active learning with uncertainty sampling to prioritize ambiguous samples for human review, improving inter-annotator agreement (measured via Cohen's kappa) [47]. Incorporating a consensus-driven annotation refinement step using conditional random fields (CRFs) can enforce spatial coherence among adjacent character labels [48]. Together, these targeted refinements—spanning advanced region proposal strategies, adaptive preprocessing, and rigorous data practices—form a comprehensive roadmap for elevating manuscript recognition performance.

B. HISTORICAL INSIGHTS

The enhanced accuracy of automated character recognition in Novgorod birch-bark manuscripts transforms the scope and scale of medieval Slavic studies. Traditionally, paleographers relied on labor-intensive manual transcription—often requiring weeks to process a single document—resulting in small, piecemeal corpora that limited quantitative linguistic and paleographic analysis [49], [50]. By achieving character detection rates through multi-scale feature fusion and geometric rectification modules, this approach enables the rapid assembly of large, machine-readable corpora spanning thousands of lines of text. Such corpora allow statistically robust investigations into lexical frequency, morphosyntactic variation, and orthographic conventions across the eleventh to fifteenth centuries [51], [52], shedding new light on dialectal shifts, the spread of diminutive suffixes, and the evolution of Slavic pronoun usage.

Beyond linguistic inquiry, reliable graphonomic features extracted from high-confidence detections support advanced scribal attribution studies [53]. Stroke-orientation histograms and curvature descriptors facilitate clustering of individual handwriting styles, making it possible to map the movement of scribes among Novgorod's artisan workshops and along regional trade routes [54]. When combined with independent dendrochronological dating of the birch bark substrata, these data have already yielded preliminary reconstructions of merchant correspondence networks, illuminating the socio-economic dynamics of urban and rural exchange in medieval Northwestern Rus'. Such insights deepen the understanding of how literacy circulated beyond ecclesiastical centers and into the hands of lay participants in early urban economies. [55], [56]

Moreover, the integration of thin-plate spline spatial transformers and denoising autoencoders has proven instrumental in resurrecting text from heavily degraded fragments [57]. Sections of legal charters, domestic records, and private letters—previously deemed illegible due to mold damage or physical distortion—can now be partially reconstructed, offering fresh evidence for the study of medieval Novgorod's legal norms, property transactions, and daily routines. This ability to recover and verify textual content bolsters interdisciplinary efforts in legal history, social anthropology, and material culture studies, allowing historians to revise and expand narratives based on newly accessible primary sources.

The methodological framework presented here—from systematic error taxonomy to active-learning–driven annotation refinement—provides a blueprint for extending these advances to other endangered manuscript traditions, including palimpsests, papyri, and parchment codices. By coupling computer vision innovations with domain-specific expertise, this work lays the groundwork for scalable digital preservation initiatives that can bring neglected or inaccessible texts into the digital humanities workflow, ensuring their preservation and enabling novel cross-cultural and diachronic scholarship.

This proposed pipeline, while developed and validated on birch-bark manuscripts, is inherently adaptable to a broad range of historical document types. Key preprocessing components—contrast-limited adaptive histogram equalization and trainable denoising autoencoders—can be retrained on substrate-specific noise characteristics, whether found in papyrus scrolls, parchment codices, or early printed paper. Geometric rectification via thin-plate spline transformations remains applicable to any non-planar medium, including bound volumes exhibiting page curvature or rolled scrolls with anisotropic deformation. Synthetic data generation strategies can be extended by extracting representative glyph or typographic elements from target corpora—such as Old Church Slavonic uncials, medieval Latin script [58], or blackletter typefaces—and compositing them under appropriate photometric and geometric distortions. Likewise, the YOLO-based detection framework and error-taxonomy-driven HPO are generalizable to different character sets and layout conventions, provided that sufficient annotated or synthetically generated training samples are available. Future studies could demonstrate transferability by applying this end-to-end approach to, for example, Dead Sea Scroll fragments [59], Renaissance watermarked papers [60], or Tokugawa-era Japanese manuscripts [61], thereby illustrating its utility for large-scale digitization and automated transcription across diverse document preservation contexts.

C. CHALLENGES

The application of YOLO-based deep learning models to birch bark manuscript recognition presents several challenges. One difficulty arises from the increasing number of character classes, which complicates training and reduces model generalization. As the number of classes grows, the risk of misclassification increases due to similarities between certain characters [62], handwriting variations, and incomplete symbols resulting from document degradation. This issue calls for advanced feature extraction techniques and possibly hierarchical classification models to improve accuracy.

Computational complexity is another factor. The real-time detection capabilities of YOLO require higher processing power when handling high-resolution images and large-scale datasets [63]. Training on diverse handwriting styles, damaged texts, and complex background noise demands extensive computational resources, making optimization strategies important [64]. External factors, such as variable lighting conditions and manuscript deterioration, also complicate recognition [65]; uneven illumination and physical damage introduce noise and additional challenges for accurate character extraction [66].

Samples subjected to accelerated aging—such as controlled ultraviolet exposure, moisture cycling [67], or enzymatic bleaching—can simulate fissuring, ink wash-out, and deep curling that commonly afflict archaeological birch-bark fragments. Performance metrics (character error rate, mAP in low-contrast regions) ought to be complemented by qualitative paleographic assessments, correlating specific failure modes with degradation patterns or regional calligraphic variants (e.g., Novgorodian ligatures or diocesan orthographies). A multimodal fusion strategy may further enhance recognition accuracy by integrating complementary data sources. Near-infrared and ultraviolet reflectance imaging often reveal subsurface ink residues, facilitating discrimination between carbon-based pigments and lignin-rich fibers [68]. When combined with philological metadata—such as frequency-weighted lexicons drawn from Old East Slavic corpora or n-gram language models—this spectral information [69] can be processed by a dual-branch convolutional or Transformer-based fusion network to enforce historical orthographic constraints and correct visually ambiguous glyphs. Such integration is expected to improve recall on faint or occluded characters while preserving precision, thereby advancing automated transcription toward the fidelity required for large-scale digital humanities and paleographic research.

D. FUTURE WORK

Future work may explore hybrid segmentation frameworks that combine classical image-processing techniques with state-of-the-art deep architectures. For example, adaptive thresholding and region-growing algorithms can provide initial proposals for character regions, which a U-Net or Mask R-CNN backbone can then refine via learned feature hierarchies and instance segmentation heads [70], [71]. Integrating attention mechanisms (e.g., SE blocks or self-attention layers) into these networks could further improve delineation of faint stroke boundaries against uneven bark textures.

The incorporation of multispectral and hyperspectral imaging holds promise for recovering severely faded or occluded glyphs. By capturing reflectance in the infrared and ultraviolet bands, it becomes possible to discriminate ink pigments from lignin-rich bark substrates and to suppress background noise [72]. Coupling these modalities with data-fusion networks—such as dual-branch CNNs or Transformer-based fusion layers—could yield enhanced input channels for downstream detection models, improving recall on low-contrast characters without sacrificing precision.

Self-supervised and domain-adaptive learning strategies offer a path toward more robust feature representations when labeled data are scarce. Contrastive frameworks like SimCLR or MoCo can pretrain encoders on large corpora of unannotated birch-bark images, enabling models to learn invariant glyph features under variable lighting and degradation [73], [74]. More recently, masked image modeling (e.g., MAE, BEiT) and vision-language pretraining have shown efficacy in historical document analysis, suggesting that combining these approaches with YOLO-style detection heads could improve generalization across new manuscripts and handwriting styles.

Drawing on error-analysis insights, several targeted refinements are recommended. First, local contrast enhancement via adaptive histogram methods or wavelet-based denoising can be dynamically tuned based on per-patch texture statistics to better separate ink from substrate noise [75]. Second, a post-recognition correction layer leveraging historical Slavic language models—such as fine-tuned BERT or GPT-3 variants trained on Old Russian corpora—can automatically rectify improbable character sequences and reduce misclassifications driven by visual ambiguity [76]. Third, contour-based refinement using active contour models (snakes) or level-set methods can be applied as a lightweight post-processing step to resolve overlapping bounding boxes and split merged symbols more accurately [77].

VI. CONCLUSION

The main finding of the study could be summarizes as it follows:

- The combined use of CLAHE and a trainable denoising autoencoder significantly improved the visibility of degraded glyphs, while the thin-plate spline rectification corrected for bark curvature. These steps produced high-quality inputs for both real and synthetic datasets, enabling robust fragment extraction and realistic data augmentation.

- Systematic comparison across SSD, Faster R-CNN, and YOLOv5–v11 revealed that YOLOv10 achieved the best trade-off between detection accuracy (mAP@0.50:0.95 = 0.85 on synthetic data) and inference speed (58 FPS). This made it the optimal backbone for downstream manuscript recognition tasks.
- An error-taxonomy-driven feedback loop for tuning anchor sizes, learning-rate schedules, and augmentation intensities increased final mAP@50 from 0.85 to 0.89. This demonstrated the value of iterative, data-driven hyperparameter refinement for small, degraded objects.
- When trained on annotated birch-bark photographs, YOLOv10 maintained strong performance—achieving mAP@50 0.75 and mAP@50–95 0.65–0.70—while losses consistently decreased. These results confirm the model's ability to generalize from synthetic to real-world manuscript conditions.
- Detailed categorization of misclassifications, missed detections, false positives, and segmentation errors highlighted key failure modes (e.g., low-contrast strokes, overlapping characters, background artifacts). These findings point to targeted strategies—such as specialized augmentations, refined annotation protocols, and alternative loss functions—to further enhance recognition accuracy.

Future experiments will quantify the impact of these innovations through ablation studies, measuring both recognition accuracy (mAP@50, character error rate) and end-to-end transcription quality (Levenshtein and Jaccard similarities). Human-in-the-loop evaluations with paleographic experts will further assess usability gains, guiding iterative improvements toward a reliable, large-scale digital archive of birch-bark manuscripts.

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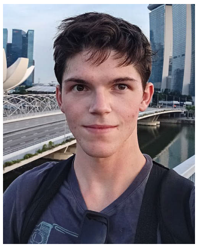

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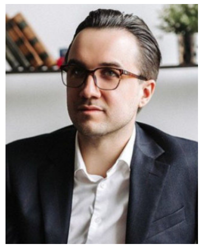

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